MH3600 Topology and Manifolds

Tutorial #1

Topics covered:

- Sets and functions
- Relations and cardinalities

Problem 1

Let X be a non-empty set. The identity mapping i_X on X is the mapping of X onto itself defined by $i_X(x) = x$ for every x. Thus i_X sends each element of X to itself; that is, it leaves fixed each element of X. Show that:

(i) $f \circ i_X = i_X \circ f = f$ for any mapping f of X into itself.

(ii) If f is one-to-one onto, so that its inverse f^{-1} exists, show that $f \circ f^{-1} = f^{-1} \circ f = i_X$.

(iii) Show further that f^{-1} is the only mapping of X into itself which has this property; that is, show that if g is a mapping of X into itself such that $f \circ g = g \circ f = i_X$, then $g = f^{-1}$.

Problem 2

Let X and Y be non-empty sets and f a mapping of X into Y. Show the following:

(i) f is one-to-one $\Leftrightarrow$ there exists a mapping g of Y into X such that $g \circ f = i_X$. Any such map is called a **retraction** of f.

(ii) f is onto $\Leftrightarrow$ there exists a mapping h of Y into X such that $f \circ h = i_Y$. Any such map is called a **section** of f.

Problem 3

Let X be a non-empty set and f a mapping of X into itself. Show that f is one-to-one onto if and only if there exists a mapping g of X into itself such that $f \circ g = g \circ f = i_X$. If there exists a mapping g with this property, then there is only one such mapping. Why?

Problem 4

Let X and Y be non-empty sets and f a mapping of X into Y. If A and B are, respectively, subsets of X and Y, show the following:

(i) $ff^{-1}(B) \subseteq B$, and $ff^{-1}(B) = B$ is true for all $B \Leftrightarrow f$ is onto;

(ii) $A \subseteq f^{-1}f(A)$, and $A = f^{-1}f(A)$ is true for all $A \Leftrightarrow f$ is one-to-one;

(iii) $f(A_1 \cap A_2) = f(A_1) \cap f(A_2)$ is true for all A_1 and $A_2 \Leftrightarrow f$ is one-to-one;

(iv) $Y \setminus f(A) \subseteq f(X \setminus A)$ is true for all $A \Leftrightarrow f$ is onto;

Problem 5

Let $\sim$ be an equivalence relation on the set X. Prove that the equivalence classes of $\sim$ form a partition of the set X.

Problem 6

Let $f : A \to B$ be a surjective function. Let us define a relation on A by setting $a_0 \sim a_1$ if

$$f(a_0) = f(a_1).$$

(i) Show that this is an equivalence relation.

(ii) Let A^* be the set of equivalence classes. Show that there is a bijective correspondence of A^* with B.

Problem 7

Suppose that A and B are two sets with order relations $<_A$ and $<_B$. The dictionary order $<$ on $A \times B$ is defined by

$$a_1 \times b_1 < a_2 \times b_2$$

if $a_1 <_A a_2$ or if $a_1 = a_2$ and $b_1 <_B b_2$. Check that the dictionary order is an order relation.

Problem 8

Show that there is a bijective correspondence of $A \times B$ with $B \times A$.

Problem 9

Show that if B is not finite and $B \subset A$, then A is not finite.

Problem 10

If A and B are finite, show that the set of all functions $f : A \longrightarrow B$ is finite.